

Capstone-II (Individual Report Outline)

1. Chapter 5: Implementation

- 1.1. Overview: Provide a brief overview of the implementation phase of your project. This should include the objectives of this phase and its importance in the overall project.
- 1.2. Development Environment:
 - 1.2.1. *Hardware:* Describe the hardware used during the implementation, including specifications and purposes.
 - 1.2.2. *Software:* List and describe the software tools, programming languages, libraries, and frameworks used.
- 1.3. System Architecture: Provide a detailed explanation of the system architecture. Include diagrams such as the system architecture diagram, flowcharts, or UML diagrams that illustrate how different components interact.
- 1.4. Implementation Details:
 - 1.4.1. *Modules Implemented:* Describe each module implemented in detail, including its functionality and purpose.
 - 1.4.2. *Algorithms/Techniques Used:* Discuss any algorithms, techniques, or methodologies applied during the implementation.
 - 1.4.3. *Challenges Faced:* Highlight any technical issues or challenges encountered during the implementation and how they were resolved.
- 1.5. Code Explanation:
 - 1.5.1. Provide snippets of key pieces of code and explain their function within the project.

2. Chapter 6: System Testing

- 2.1. Overview of Testing Process: Briefly describe the testing process, its objectives, and its significance in ensuring the project meets its requirements.
- 2.2. Testing Environment:
 - 2.2.1. *Hardware:* List the hardware used during testing.
 - 2.2.2. *Software:* Describe the software environment, including any testing tools or frameworks used.
- 2.3. Test Cases:
 - 2.3.1. *Test Scenarios:* Outline the scenarios tested, why they were chosen, and what they aim to verify.
 - 2.3.2. *Test Cases Description:* Provide detailed descriptions of individual test cases, including:
 - 2.3.2.1. *Objective:* What is the test case designed to verify?
 - 2.3.2.2. *Input:* What inputs are provided for this test case?
 - 2.3.2.3. *Expected Outcome:* What is the expected result?
 - 2.3.2.4. *Actual Outcome:* Document the actual results and whether they matched expectations.
- 2.4. Test Results:

- 2.4.1. *Summary of Test Results:* Provide a table or summary showing which tests passed and which failed.
- 2.4.2. *Discussion of Failures:* If any tests failed, discuss why and how these issues were addressed.
- 2.5. *Performance Testing:* Provide details of any performance tests conducted, including objectives, methodology, and results.
- 2.6. *Validation and Verification:* Discuss the methods used to ensure the system meets all functional and non-functional requirements.

3. Chapter 7: Conclusion and Critical Evaluation

- 3.1. *Summary of Work Done:* Recap the key elements of the project, from the objectives set out at the beginning to the implementation and testing phases.
- 3.2. *Critical Appraisal of the Project:*
 - 3.2.1. *Strengths:* Discuss the strengths of your project, including what worked well and why.
 - 3.2.2. *Weaknesses:* Reflect on the weaknesses or limitations of your project, including any aspects that did not work as expected.
 - 3.2.3. *Challenges Faced:* Recap the challenges faced during the project and how you overcame them.
- 3.3. *Future Improvements:* Suggest any potential improvements or future work that could enhance the project, including additional features, optimizations, or alternative approaches.
- 3.4. *Personal Reflection:* Reflect on what you learned throughout the project. This can include technical skills, project management, teamwork, or any other relevant experiences.

Note: For each group project, **one hard-bound report (hard copy)** must be submitted, including all chapters from **Chapter 1 to Chapter 7**, along with properly formatted **appendix** and **reference** sections.